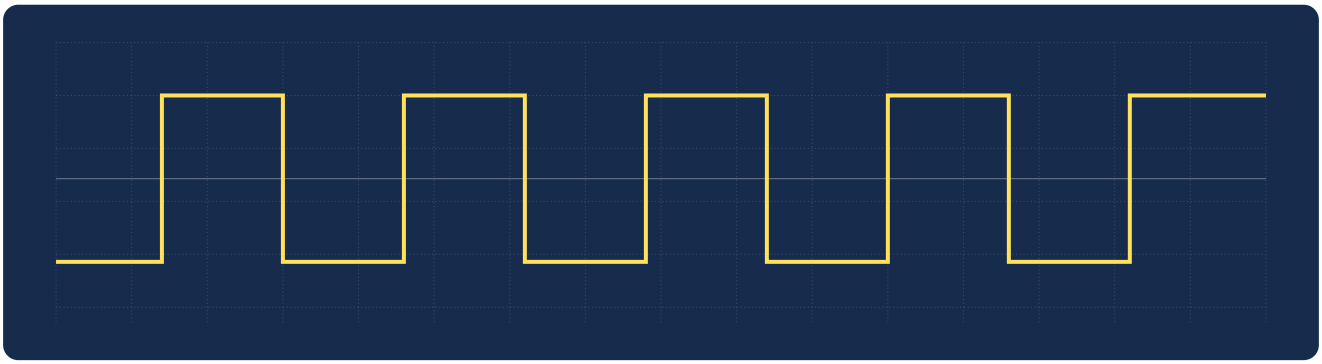
DSO138
Oscilloscope

User manual

Classic board | C++ firmware



4 keys
Run, hold and edit

9 ranges
10 mV/div to 5 V/div

18 timebases
10 μs/div to 5 s/div

Find what you need

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Manual revision 1.3 | 2 November 2026

Firmware source: unreleased revision (UID-configured build required)

Intended hardware: PCB 109-13800-00C, STM32F103C8T6, ILI9341 display.

This guide describes the implemented controls and functions. Nominal readings are not a calibration certificate.

1 Quick start

Board and power

This guide is for the identified DSO138 board, PCB 109-13800-00C, with STM32F103C8T6 and ILI9341 display. Use the board's normal 9 V DC input as marked on the board. For bench work, use a low-voltage isolated supply and connect signal ground to the scope ground. Use only isolated low-voltage bench signals; never connect the scope input or ground directly to mains. Do not infer a safe maximum input from this guide; observe the board markings and its analog input limits.

This operating guide assumes a correctly configured firmware image. The current unreleased source revision requires each owner to set the MCU UID and board identity before normal startup; see the repository's Getting started guide.

Try the built-in signal

1. Connect the onboard test output to the signal input.
2. Set the physical coupling switch to DC, base sensitivity to 1 V/div, and multiplier to $\times 1$.
3. Press **SEL** until Timebase is selected. Use **+** / **-** to select 1 ms/div.
4. Select Trigger mode and choose AUTO with **+** / **-**.
5. In an ordinary field such as Timebase, press **OK** if the scope is in HOLD. Look for RUN and a repeating square wave.
6. Press **OK** again to hold the record for inspection.

The observed board showed about 3.3 vertical divisions and a frequency near 1 kHz in this setup. This is a functional check, not a guaranteed voltage, frequency standard, or calibration reference.

On power-up, saved valid settings return in HOLD; press **OK** to run. If there are no saved settings, the enabled board profile requests Auto run after the physical switches settle. Waveform recall is always a separate, explicit action. Allow the display to show a record before judging a signal.

The three slide switches set the analog input. Their positions physically select coupling and sensitivity. Software settings cannot move them. After changing a switch, wait for the on-screen status to leave `WAIT INPUT` or `Switches settling` before using a new record.

2 Buttons and physical switches

Control	Action
SEL	Move to the next setting field. After Domain, the sequence returns to Timebase.
+ , -	Change a value or cycle a choice. For numeric settings, holding a button repeats after about 400 ms, then at about 80 ms intervals. Discrete choices do not repeat.
OK	Run or hold in ordinary fields; rearm Single when appropriate. In Storage or Calibration, the first press stops acquisition and a later press executes the selected action. Release between presses.
RESET	Hardware reset. Unsaved edits are lost; the last saved configuration normally returns in HOLD. A reset during flash erase/program can lose saved data; see Saving.

The controls accept one button action at a time. If buttons are pressed together, release them before trying again. A completed Single capture or a fault may need an acknowledgement before the next action; wait for the status to settle, release, then press OK again when the screen indicates the action is ready.

Base sensitivity	Multiplier $\times 1$	Multiplier $\times 2$	Multiplier $\times 5$
10 mV/div	10 mV/div	20 mV/div	50 mV/div
100 mV/div	100 mV/div	200 mV/div	500 mV/div
1 V/div	1 V/div	2 V/div	5 V/div

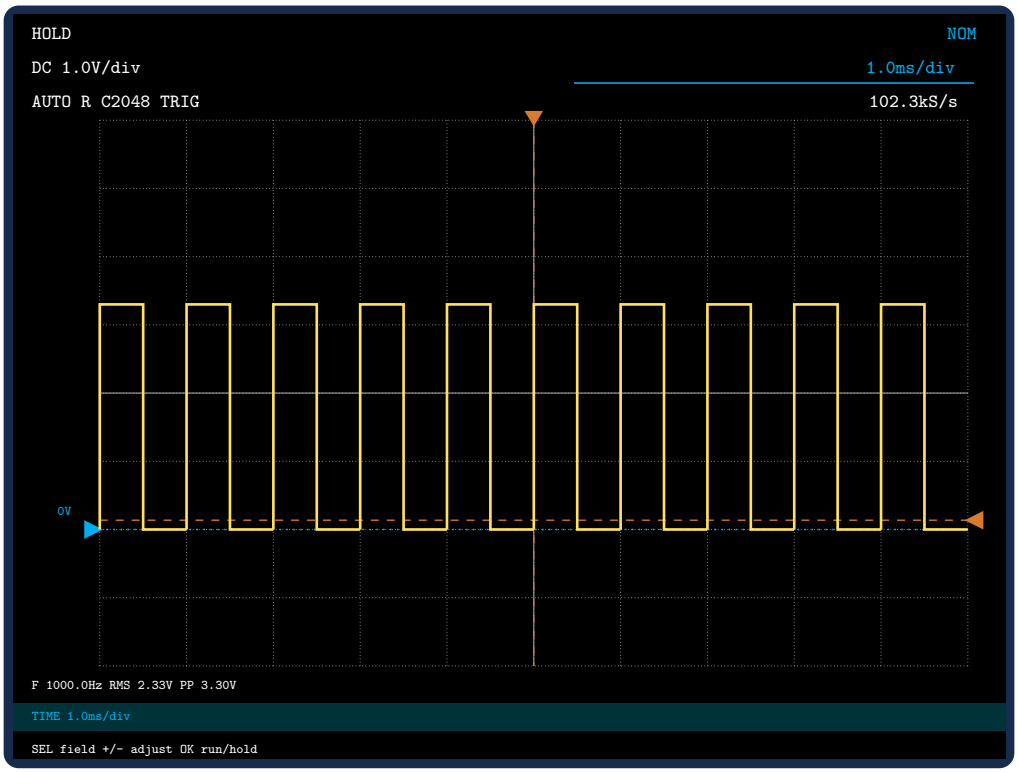
The separate coupling switch selects DC, AC, or GND.

The table gives the nine nominal ranges. The separate coupling switch selects DC (shows the input offset), AC (blocks DC), or GND (selects the internal ground path). These are front-panel sensitivity labels. They do not confirm the input voltage or a probe's attenuation. If using a $\times 10$ probe, account for that factor yourself; there is no probe-attenuation menu.

Trigger activity LED

D3 gives a brief, about 100 ms indication for an accepted trigger crossing; another crossing restarts the pulse, so close events may look continuously lit. Auto free-run timeout and recall/HOLD redraw do not light it. A Single pulse can finish through automatic HOLD; faults leave it off. This software cue is not a precise timing strobe and has not been tested on hardware.

3 Reading the screen



Illustrative layout, not a screenshot or a measured record.

The display status tells you what the instrument is doing: `RUN` or `ARM` indicates active acquisition or arming, `HOLD` indicates a stopped view, and `WAIT INPUT` or `WAIT BACK` means it is waiting for stable switches or an acquisition backend. The selected Domain controls whether the display shows the scope trace or the spectrum of a completed record.

The plot has 10 horizontal and 8 vertical divisions. The yellow trace/envelope shows available samples; drawing does not create extra samples. Orange marks the trigger position and level; cyan marks the zero reference and selected control. Focus labels beginning `NEXT` show the value to be used by the next capture, while a held trace continues to show its frozen capture settings. The reference line is labeled `0V` with a usable voltage profile, or `REF` when values are raw ADC codes.

The top labels show coupling, sensitivity, timebase, trigger mode/edge/level and sample spacing or rate. A held waveform keeps the range, coupling, timing and calibration provenance from its capture. If a slide switch changes, the old held trace is not rescaled to the new analog setting; the display may say `OLD range/coupling held`. Start a fresh capture to inspect the new setting.

Reading the grid: 3.3 vertical divisions at 1 V/div represent about 3.3 V peak-to-peak. One full cycle spanning one horizontal division at 1 ms/div represents a 1 ms period, or about 1 kHz. Apply an external probe's attenuation factor separately.

The quality tag describes voltage conversion: `NOM` uses nominal board coefficients, `CAL` means a measured profile is installed, and `RAW` means no voltage conversion is available. None of these labels by itself certifies measurement accuracy. Clipping warnings mean amplitude and frequency results are invalid for that record.

4 Menu and settings

Press SEL to move through the thirteen fields. Use + and - to adjust; use OK as shown. The focused field is shown near the bottom of the screen.

Focus	+ / -	OK
Timebase	Select a catalogue setting	Run/Hold
Trigger mode	AUTO, NORMAL, SINGLE	Run/Hold; Single rearm
Trigger edge	Rising or falling	Run/Hold; Single rearm
Trigger level	ADC threshold, 16-code steps	Run/Hold; Single rearm
Trigger position	0 to 100%, 2.5% steps	Run/Hold; Single rearm
Vertical position	Move reference, 2-pixel steps	Run/Hold; Single rearm
Horizontal position	Pan, 16-sample steps	Run/Hold; Single rearm
Measurements	Overview, Amplitude, RMS/Mean, Freq	Run/Hold
Storage	Choose one of six actions	Stop, then execute
Calibration reference	Set signed reference value	Stop, then cycle step
Calibration tolerance	Set allowed error	Stop, then cycle step
Calibration	Begin, Accept Ref, Install, Abort	Stop, then execute
Domain	Spectrum or Scope	Run/Hold

For Storage and Calibration, if acquisition is running, the first OK changes to HOLD. Release the button, then press OK again to carry out the selected action. In Reference and Tolerance, OK in HOLD cycles the decimal edit step; + and - change the value by that step. In Domain, + selects Spectrum and - selects Scope; the choice is session-only and the instrument starts in Scope after boot. `SAVE CONFIG` still saves the eight scope settings listed under Storage; it does not save the Domain choice. Reference/tolerance editors are not saved by that action.

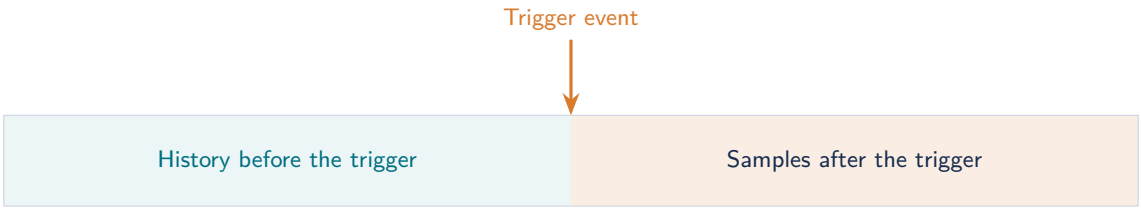
5 Triggering and viewing

- **AUTO:** publishes a record when a trigger edge qualifies, or after its timeout if no qualifying edge arrives. Use it for a general live view and for the built-in signal check.
- **NORMAL:** waits for the selected edge at the trigger level. If the signal never crosses the level in the selected direction, it can keep waiting.
- **SINGLE:** captures one complete record and stops. Press OK to arm another capture. If the completed record or a fault has not yet been acknowledged, wait for the screen state, release, and press again.

Choose rising or falling edge in Trigger edge. Trigger level changes in 16 ADC-code steps; it is a code threshold, not a calibrated voltage setting. Trigger position moves from 0% to 100% of the record in 2.5% steps, selecting how much history lies before the trigger in ordinary triggered captures.

Timebase, trigger mode, edge, level and trigger position affect capture. Changing them during a run invalidates the incomplete record. On a held or recalled trace, timebase changes zoom the existing samples; they do not resample the waveform. Areas outside the captured record remain empty. Vertical + moves the reference down; - moves it up. Horizontal + pans toward later source samples; - pans toward earlier samples, in 16-sample steps. These view adjustments do not make a new capture.

Example: trigger position at 50%



Record position before any view pan or timebase zoom.

Use HOLD to inspect a stable record, then RUN to resume. With AC coupling, the instrument cannot show the DC component removed by the input circuit. With GND coupling, the trace shows the grounded input path, not an external signal.

6 Measurements and validity

Measurements cover the complete finite record, not just the visible portion or a hand-selected cycle. The four measurement pages are Overview, Amplitude, RMS/Mean and Freq.

Screen item	Meaning
MIN, MAX	Lowest and highest sampled values in the record.
PP	Peak-to-peak, the difference between maximum and minimum.
PK	Peak magnitude from the voltage zero; it is not necessarily half of PP when a DC offset is present.
RMS	Total RMS about the calibration zero, including DC.
AC, AC-RMS	RMS after subtracting the record mean.
MEAN, AVG	Arithmetic mean of the record.
F	Detected repetition rate, when valid.

With `RAW`, the display reports ADC codes (marked c) rather than input volts. `NOM` uses the board's nominal conversion coefficients. `CAL` means an installed measured DC profile is being used. A displayed voltage is a converted estimate, not an accuracy guarantee. AC-coupled mean and total RMS describe the AC-coupled signal; they cannot restore removed DC.

Frequency needs at least three qualified complete periods. A clear trace can still show `-- CYCLES` when it is too short. Other reasons can include low amplitude, instability, clipping, grounding or a rate outside estimator limits. Frequency is observed repetition, not proof of the signal's fundamental frequency. Aliasing can produce a plausible but incorrect value.

An ADC rail hit suppresses valid amplitude and frequency claims; the scope view may show `ADC CLIP`, `RAW ADC CLIP`, or `F --CLIP; volts invalid`. A missing voltage profile gives `RAW: no volts`. Stop and correct the input/range before relying on another record. Clipping can add harmonics to the Spectrum view; it has no separate clipping warning, so check the same record in Scope view. As a mathematical illustration only, an ideal 0-to-3.3 V, 50% duty square wave has 3.3 V PP, 1.65 V mean, 2.33 V total RMS and 1.65 V AC RMS. These calculated values are not a test or specification of the onboard output.

7 Spectrum view

Spectrum view analyzes 512 consecutive samples from a completed, frozen record. The calculation uses the record's known timer rate and the same sample block throughout. With RUN active, a new completed capture produces a refreshed spectrum; analysis runs after capture, not while the ADC is sampling. HOLD and recalled records can also be viewed. A record must contain at least 512 samples and have known timing.

The analyzer subtracts the block mean and applies a fixed periodic Hann window before the 512-point FFT. Mean removal suppresses DC, so a constant or zero input has no positive-frequency peak and the readout shows `Peak --`. The Hann window reduces leakage from tones between frequency bins, while spreading a tone over nearby bins. The peak readout uses the strongest positive-frequency bin without interpolation, so its frequency is quantized in steps of $\Delta f = F_s/512$.

Display item	Meaning
Horizontal axis	From 0 Hz to the Nyquist frequency, $F_s/2$. The label `df` gives the bin spacing $F_s/512$. F_s is the configured sample rate derived from the timer plan, not an independently measured clock or accuracy specification.
Vertical axis	Fixed 80 dB span from +10 to -70 dBFS, at 10 dB/div. The 0 dBFS marker is 22 pixels below the plot top.
`Pk` readout	Strongest positive-frequency bin, in hertz, followed by its level in dBFS.

dBFS uses a peak reference of 2048 ADC codes: 0 dBFS corresponds to a 2048-code sinusoidal peak in the input-code scale. It is not a voltage reading, and the Spectrum view does not use the installed voltage calibration. Length, window, frequency span and vertical scale are fixed.

Nyquist is the highest frequency represented without folding for the sampled block. Higher-frequency energy can alias into the displayed band and appear as a plausible lower-frequency peak; the Hann window does not prevent aliasing. Use suitable analog bandwidth limits ahead of the ADC when the signal may contain energy above Nyquist. ADC clipping can create harmonics that look like real signal components. Spectrum view has no clipping warning; check the same record in Scope view for its ADC clipping status.

Accessing the spectrum

Use **SEL** to focus Domain, then press + for Spectrum or - for Scope. In RUN, each new complete record is analyzed after capture; in HOLD, the current or recalled record remains available. The Domain choice is not stored by `SAVE CONFIG` and resets to Scope after restart.

8 Saving, loading and recalling

Storage offers six actions in the Storage field. Select one with + / -; waveform saving requires a complete, fully processed held record. A valid recalled record may also be saved. If running, press OK, release, then press it again to execute. Wait for SAVED after a save before changing power or resetting.

Displayed action	What it does
SAVE WAVE	Store the latest complete waveform and current eight scope settings in the single waveform slot, with captured timing, physical and calibration metadata.
RECALL WAVE	Restore the saved settings and record together as a held view. Recall is explicit; it does not install its calibration into the live bank.
SAVE CONFIG	Save the eight scope settings (timebase, trigger mode/edge/level/position, vertical/horizontal position and measurement page). It preserves the waveform slot.
LOAD CONFIG	Restore saved UI settings. The restored settings start in HOLD.
SAVE CAL	Persist the currently installed calibration bank.
RESTORE CAL	Restore the saved calibration bank into the live profile. A valid calibration bank is also restored at boot.

To save a trace: obtain a complete record, press OK to HOLD if necessary, select SAVE WAVE, then execute with a fresh OK. This stores the waveform and its current scope settings together. To recall: select RECALL WAVE in HOLD and execute; the saved record and settings return together in HOLD. To save settings alone without replacing the waveform slot, select SAVE CONFIG. Reference and tolerance editor values are not part of the eight saved scope settings; calibration persistence is separate through SAVE CAL.

> **Single-bank flash warning**
>
> Saved records occupy one 2 KiB storage region in the device flash. A reset or power loss during flash erase/program can lose the waveform, settings or calibration. Do not interrupt a save; wait for SAVED. Completed SAVE CONFIG and SAVE CAL operations normally preserve the waveform; SAVE WAVE replaces it. The lower-bank area used by older firmware now holds the application, and migration of older saved data is not guaranteed.

Flash endurance is finite. Avoid repeated unnecessary saves. Physical interruption recovery and endurance have not been fully qualified. HOLD TO SAVE means first obtain and hold a complete processed record. A recalled waveform cannot be used as an applied calibration reference.

Records saved with one of the removed 10 s/div through 500 s/div timebase indices are not recalled by this firmware. Opening such a record does not erase or rewrite it. An explicit save of supported settings, a waveform, or calibration can replace the single storage slot.

9 Optional DC calibration

Leave the nominal profile in place unless you have suitable, stable reference-voltage equipment. The onboard test output is not a calibration standard. Calibration converts ADC codes to voltage for DC measurements; it does not qualify AC response, bandwidth, the clock or probe attenuation.

The procedure uses two DC references to fit a profile and a separate third DC reference to check it. Each must be a fresh held live record with unchanged physical switch settings, at least 64 samples, spread no greater than 32 ADC codes, and no rail samples. The two fitting records must differ by at least 64 ADC codes. The third must lie at least 16 codes inside both fitted endpoints and pass the selected tolerance. A reference value is entered for each record. Enter values in microvolts (μV): $1{,}000{,}000\,\mu\mathrm{V} = 1\,\mathrm{V}$. These editor values are not input ratings.

To acquire each reference, leave the calibration fields with **SEL**, select an ordinary field such as Timebase, then use **OK** for RUN and HOLD. **OK** inside a calibration field never starts acquisition.

1. Set DC coupling and one fixed range. In HOLD, choose `BEGIN` and execute. Apply reference 1, set its known value in Calibration reference, RUN and acquire, then HOLD and choose `ACCEPT REF`.
2. Apply a different reference 2. Update Calibration reference to its known value, acquire a fresh record, HOLD, then choose `ACCEPT REF` again.
3. Apply independent reference 3. Update Calibration reference to its known value and set Calibration tolerance to the allowed error. Acquire a fresh record, HOLD, and choose `ACCEPT REF`. The session should show `VERIFIED`.
4. Choose `INSTALL` and execute. The profile is now installed in RAM for the matching DC range. To keep it across restart, select `SAVE CAL` in Storage and execute separately.
5. Choose `ABORT` to discard a pending candidate. Abort leaves any already installed profile in place.

In Reference and Tolerance, **OK** cycles the decimal step (powers of ten); it does not move a cursor digit by digit. **+** / **-** changes the numeric value by that step. Replace an unsuitable held record with a fresh one. A switch change, failed two-point fit or failed third-point check requires `BEGIN` and a new session. Installation and saving do not certify hardware accuracy; an external comparison is still required.

10 Troubleshooting

Screen status or symptom	What to do
No trace or an empty plot	Choose AUTO, return to RUN and wait for a record. Set vertical and horizontal position to zero; a panned or zoomed held view can lie outside the captured data.
`WAIT INPUT`, `Switches settling`	Check the physical slide-switch positions and allow them to settle.
`WAIT TRIG`	In Normal or Single, check that the input crosses the selected trigger level and edge.
`FFT NEEDS FROZEN FRAME`	Complete a capture and hold it before selecting Spectrum.
`FFT NEEDS HELD UNIFORM BLOCK`, `FFT NEEDS 512 SAMPLES`	Hold a complete record with at least 512 uniformly timed samples.
`FFT TIMING UNKNOWN`	Select a timebase with a valid timer plan, then acquire a fresh record.
`Peak --`	The mean-removed block has no nonzero positive-frequency bin. A DC-only input is removed before analysis.
`-- CYCLES`	Select a wider time span so more periods fit in the fixed record. For the onboard signal, try 1 ms/div.
`RAW: no volts`	Voltage calibration is unavailable. Read ADC codes only or restore/use an appropriate profile.
`ADC CLIP`, `RAW ADC CLIP`	Reduce the input or choose a suitable physical range, then capture again. Clipping can add false harmonics in Spectrum view.
`OLD range/coupling held`	The held record belongs to the previous switch setting. Run a fresh capture.
`HOLD TO SAVE`	Wait in HOLD for a complete, fully processed record, then retry.
`EMPTY`, `READ ONLY`, `DISABLED`	The requested storage item is absent or the action is unavailable; check selection and status.
`ERROR`, `Save failed; kept; OK retry`	The previous record remains held when available. Wait, then retry once with a fresh **OK**; if it repeats, stop using storage and restart normally.
`FAULT`	Wait for the fault state, release all buttons, then use a fresh **OK** retry if offered.

If a button appears ignored, press only one at a time and release it fully before the next action. In Storage or Calibration, remember that the first **OK** may only stop acquisition. Use a second press after release to execute. Do not reset during a save.

11 Reference and scope

Timebase catalogue (time per horizontal division):

Unit	Values
Microseconds (μs/div)	10, 20, 50, 100, 200, 500
Milliseconds (ms/div)	1, 2, 5, 10, 20, 50, 100, 200, 500
Seconds (s/div)	1, 2, 5

This firmware supports 18 triggered settings from 10 μs/div to 5 s/div. The original DSO138 and other firmware may describe longer settings; they are not available in this firmware. The configured fast acquisition rate reaches 500 ksample/s; this is a timer plan, not an independently measured sample rate, jitter or timebase accuracy. The horizontal scale, display refresh and acquisition rate are different quantities.

Voltage and probe limits. Nominal range labels and `NOM` conversion use board coefficients; they are not a production calibration. `CAL` indicates a measured profile is installed, but does not replace independent verification. `RAW` provides codes, not input volts. No probe attenuation is selected by firmware; multiply displayed input estimates by the probe factor when applicable. Do not treat the onboard signal as an absolute reference. No maximum safe input is specified here: follow the board's own input markings and limits.

Firmware identity and qualification

The operating and qualification observations in this manual refer to the previously deployed b73 firmware image on the identified board profile, with nominal gain-2 coefficients and a grounded-zero code; its reported build image SHA-256 begins `b73cdcb1c644`. Public firmware source release 1.0.1 adds the trigger activity LED described under Controls. Its LED pulse, acquisition handoff and startup/fault sequencing passed software checks; 1.0.1 has not been flashed or tested on hardware. The current unreleased source revision changes saved-data layout to a single flash bank and adds a software FFT display; these changes have not been flashed or tested on hardware. This change does not complete physical production qualification. Independent voltage, frequency, bandwidth, loaded timing, FFT analog accuracy, alias response and long-term storage qualification remain open. Use the instrument as a functional oscilloscope with these limits in mind, not as a calibrated laboratory instrument.

Manual revision 1.3: 2 November 2026. Operator reference sources: `firmware/App/Src/dso_ui_model.cpp`, `firmware/App/Src/dso_ui_controller.cpp`, `firmware/App/Src/dso_scaling.cpp`, `firmware/App/Src/dso_spectrum.cpp`, and `firmware/App/Inc/dso_timebase_presets.hpp`.